

Access Is Not Use: Auditing Causal Bypass of Force in Behavior-Cloned Vision-Language-Action Policies

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Abstract—Vision-language-action (VLA) policies increasingly receive physical condition signals—wrist wrench, contact, vacuum-seal state—yet whether a trained policy *causally uses* them is unmeasured, and standard validation cannot tell: we show that a policy that uses force and one that routes around it can train to validation losses equal in the fourth decimal place. On a real-robot suction case-picking task where the cases carry battery cells—a 15 N contact-force limit, and a critical transition (stop descending on contact) observable only through force—a naively fine-tuned VLA receiving the full wrench presses through contact in every trial and never stops on its own. We present a recipe that makes the policy use force where it matters: a condition vector $\hat{c}$ computed from signals the policy already receives, injected through FiLM with the raw wrench masked—a grounded bottleneck that adds routing, not information—trained on press-retreat demonstrations that press to a varied force target and retreat before the seal. Forcing $\hat{c}$ at inference measures its causal authority, and a controlled ablation shows each ingredient is necessary: unmasked, the pathway trains causally dead (*bypass*); ungrounded, the policy is the worst in the study. The naive policy’s force response is a template that fades beyond the demonstrated range; the conditioned policy’s grows monotonically, stopping and retreating at forces beyond its training data. On the robot, every conditioned pick stops on its own—median contact force 2.5 N across three stack heights—where the naive policy is externally aborted in every attempt, and live counterfactual probes reproduce the offline crossover on the physical system. **[TODO: scale sentence ($\pi_{0.5}/\pi_0$) pending the offline re-run]**

Index Terms—imitation learning, vision-language-action models, force control, causal confusion, contact-rich manipulation

I. INTRODUCTION

Behavior-cloned vision-language-action (VLA) policies are rapidly acquiring physical condition signals. Wrist force/torque, contact events, and task-state bits now routinely appear in the observation vector, and a growing line of work fuses force into the policy backbone for contact-rich manipulation [1]–[4]. All of this work shares a validation recipe—add the signal, retrain, and validate by task success (occasionally alongside rollout force statistics [2], [3])—and none of it measures whether the trained policy *causally uses* the signal when it matters: whether, holding everything else fixed, changing the force input changes the action. Imitation learning offers no such guarantee. The objective asks the

policy to reproduce *what* the expert did, not *why*: a clone can imitate the behavior convincingly while being driven by entirely different cues than the expert was. Imitation pins down the trajectory, not the mechanism—equally faithful clones can act from opposite causes, and the loss cannot tell them apart [5], [6].

The consequence is easiest to see where force is the only signal that can carry the task. We study real-robot suction case picking with a single head-mounted camera and no wrist camera: the arm must stop descending the moment the suction cup meets the case, and stack heights differing by a few millimeters—the entire success margin—are visually near-indistinguishable at head-camera range. The wrench is in the state the policy receives, and every demonstration exhibits the stop. Yet a naively fine-tuned VLA—full wrench in its state, hereafter the *naive policy*—presses through contact in every evaluation trial and never initiates a stop of its own: each run ends only when an external safety layer, separate from the policy, trips at its 15 N force limit—set to protect the battery cells the cases carry (15.4–18.7 N measured at abort)—the policy still commanding descent. The task turns on a single transition—stop when force rises—and it is precisely this transition that the policy fails to bind to force.

This failure leaves no trace in training. A policy that binds its actions to force and a policy that routes around it can reach the same validation loss—in our experiments, equal to the *fourth decimal place*—so no loss value can certify a fix. We call this routing-around *bypass* and exhibit it directly in Sec. V. A fix must therefore hold *by construction*: leave the policy no route around the signal, and make its use of the signal directly measurable.

Our prescription does both. A condition vector $\hat{c}$ (contact, normal force, seal) is *computed* from signals the naive policy already receives and injected through FiLM modulation [7], and the raw wrench is masked from the state, so $\hat{c}$ becomes the *only* pathway from force to action—no route around. We call this design a *grounded bottleneck*—a bottleneck because $\hat{c}$ is the sole route from force to action, grounded because its content is computed from measured signals—and the policy trained behind it the *conditioned policy*. By construction the

conditioned policy receives no information the naive policy lacks; the design changes routing, not content. And because the pathway is explicit, forcing $\hat{c}$ at inference directly measures its *causal authority*—the fraction of commanded descent the condition can cancel. The demonstrations complete the recipe: each press-retreat episode presses to a randomized force target and retreats before any seal event, so the force at which the expert stops varies from episode to episode—in the data, depth, appearance, and timing no longer predict the stop; only the force reading does. The same interface that fixes the policy is the instrument that audits it (Sec. IV).

The resulting policy is not uniformly better—it is different exactly where the task is decided. In a matched comparison—same data, same initialization, imitation error indistinguishable—the naive policy *does* respond to force at the demonstrated magnitudes, as strongly as the conditioned policy or more so. But its response is a *template*: sized to the demonstrated forces, it fades or saturates as force rises past them, never strengthening into a stop. The conditioned policy’s response instead grows with force, to the point of stopping and backing away at forces beyond anything in the training data (Sec. V). The mechanism carries to the real closed loop: while the naive policy drives contact force past the 15 N safety limit on every attempt, the conditioned policy stops on its own in every pick—across three stack heights, at a median contact force of 2.5 N—and live counterfactual probes reproduce the offline crossover on the physical system (Sec. V-E).

Our contributions:

- **From force runaway to force regulation, with the same demonstrations.** A grounded FiLM bottleneck— $\hat{c}$ computed from signals the policy already receives, injected through FiLM, the raw wrench masked—turns the same press-retreat demonstrations into a policy that brakes monotonically with force instead of reproducing a saturating template: on the robot it stops under the 15 N abort limit in every pick (median contact force 2.5 N) on a transition the naive policy fails uniformly (Sec. IV, V-E).
- **An instrument for causal authority.** The bottleneck is measurable by construction: a counterfactual probe battery—condition forcing, state transplants with pathway decomposition, dose-response sweeps, closed-loop press simulation, and on-robot frozen-observation counterfactuals—quantifies moment-resolved causal use and transfers to the real system (Sec. IV-C).
- **Robustness beyond the demonstrations.** The conditioned policy’s stop is bound to force itself, not to force’s correlates—under intervention, its response runs entirely through the conditioning pathway—and this buys robustness on the two axes we test: braking grows monotonically at forces beyond anything demonstrated, stopping and retreating where the naive policy’s template saturates and presses through, and picks succeed 3/3 at a stack height absent from the demonstrations. The failure class the recipe corrects persists unchanged at 3.6B scale (Sec. V, V-E).

In one sentence: *a policy that stops when it feels what the demonstrations felt and a policy that stops harder the harder it is pressed are indistinguishable in loss and similar at the demonstrated forces—they differ exactly where the task is decided, and only demonstrations in which force alone predicts the stop, behind a grounded, masked FiLM bottleneck, yield the second.*

II. RELATED WORK

Force- and tactile-conditioned VLA policies. ForceVLA fuses wrist force as dedicated tokens through a mixture-of-experts backbone [1]; TA-VLA injects torque adapters into the action decoder [8]; PhaForce schedules policy phases on contact and force signals [2]; ForceFlow [3], FARM [9], and FILIC [4] condition diffusion or impedance-loop policies on force and tactile input; TacForeSight learns force-conditioned tactile prediction [10]. These works *design for* force use and validate it with task success and retraining ablations, sometimes alongside rollout force statistics [2], [3]. None of these measurements establishes causal use at the decision moment: an ablation compares two differently-trained functions, rollout statistics are observational, and success can be carried by correlates of force rather than force itself. The nearest analyses stop short of the force input: TA-VLA perturbs input tokens of its *baseline* to site the adapters and reports correlational representation similarity [8]; Tactile-VLA’s dose-response intervenes on the *instruction*, not the force stream [11]; ForceFlow masks its trained policy’s *visual* input and reads out task success [3]. We do not claim these systems bypass their force inputs—their tasks may contain force information that vision cannot substitute, leaving no shortcut to take. Our claim is narrower and falsifiable: *when* the demonstrations correlate the condition with cheaper cues, a policy with a dedicated force pathway can train to indistinguishable loss while routing around that pathway (we exhibit exactly this with our unmasked control), and no metric in the standard validation stack detects it. We contribute the missing measurement: an inference-time intervention on the trained policy’s force input, with the action as the readout.

Intermediate condition representations. Injecting a compact, task-level representation between observation and action is a recurring design: RT-Affordance conditions on predicted affordance plans [12], and FiLM modulation [7] is the standard mechanism for conditioning a backbone on side information. Our $\hat{c}$ shares the shape but differs in role: it is *computed* from signals the policy already receives—not a learned head that could itself be bypassed—and it is paired with a mask that removes the raw signal, turning the representation from a convenience into a measurable bottleneck.

Causal confusion and shortcut learning. That imitators latch onto spurious correlates is well established: causal confusion in IL [5], copycat behavior on observation histories [13], and shortcut learning in deep networks broadly [6], [14]. Recent work documents shortcuts in generalist robot policies [15] and uses targeted interventions to expose or

correct them [16], [17]. The symptoms are on record in force-conditioned policies themselves: FuSe observes fine-tuned generalists ignoring newly added sensors [18], and FM-VLA diagnoses a history-length shortcut in its own force stream and patches it with augmentation—verifying the fix, again, only by task success [19]. We instantiate this literature where it has not been applied—force conditioning in fine-tuned VLAs—and add two elements: a quantitative demonstration that *training loss carries zero signal* about which input the policy is bound to (equal to the fourth decimal in two independent model pairs), and the usage/form dissociation: the data’s correlation structure decides *whether* a signal is used, while the conditioning architecture decides *what form* its use takes under extrapolation.

Privileged supervision and interventional evaluation. Privileged-teacher distillation [20], [21] assumes access to a signal worth distilling; our probes answer the prior question of whether the deployed policy uses the signal at all. The closest interventional precedent is ForceSight, which ablates predicted force *goals* at its controller under matched initial conditions [22]; the intervention sits on the output side of a model that takes no force input, and reads out end-of-episode success rather than the action. Our press-retreat prescription is a data-side intervention in the spirit of decorrelation fixes for causal confusion [5], but designed from a physical argument (Sec. VI) that reactive gating is arithmetically insufficient on stiff contact, so the demonstrations must teach anticipatory approach and retreat.

III. TASK, SETTING, AND THE FAILURE

Task and platform. We study suction case picking on a Dexmate Vega-1p, a dual-arm mobile manipulator: the arm descends onto a stack of rigid cases, must stop when the suction cup meets the top case, hold until vacuum seal is confirmed, and lift (Fig. ??). The cases carry battery cells, which tolerate little compression—over-pressing risks the product, not just the pick—so deployment enforces a 15 N contact-force limit. The margin is tight in both directions: too little press and the seal fails; a few millimeters too much and contact force climbs past the limit, since the stack is stiff (~ 5 N/mm, Sec. VI). The policy controls a 7-D end-effector action (3-D position and 3-D rotation increments plus a suction command) at 15 Hz, executed in chunks of 5 actions (~ 330 ms between observations).

Observations. The policy observes a single head-mounted camera (RGB and colorized depth; no wrist camera) and a 15-D state: end-effector position and quaternion, suction command, vacuum-seal bit, and the 6-D wrist wrench. The setting is a deliberately clean diagnostic: at head-camera range, case-stack heights differing by millimeters—the entire success margin—are visually near-indistinguishable (Fig. ??, right), while the wrench and seal bit signal the contact transition directly. Every condition the task depends on is *in the state the policy receives*; the question is whether the policy binds its behavior to it.

Demonstrations. We collect 100 training episodes (19,445 frames) and hold out 6 validation episodes (1,169 frames) with a scripted expert designed to *decorrelate* force from its habitual correlates: each episode descends with suction on, presses to a randomized force target drawn from $[8, 15]$ N, retreats 5–10 mm from the contact plane, then hovers and lifts immediately on seal. Episodes are collected at two different stack heights, so the contact plane itself moves across the dataset. Every episode thus demonstrates the counterfactual the task requires: force rises $\rightarrow$ stop $\rightarrow$ move back up, *before* any seal event, at a force level that varies independently of depth. This is the data-side prescription whose rationale we develop in Sec. VI; here it matters that the naive policy and all conditioned variants are trained on exactly these episodes.

The failure. Deployed on the robot, the naive policy—a SmolVLA [23] fine-tuned on these demonstrations, with the full wrench and seal bit in its state—fails every evaluation trial the same way: it descends, makes contact, and keeps pressing—never initiating a stop of its own—until the external safety layer aborts the run (15.4–18.7 N measured at abort). The transition that the demonstrations exhibit in every single episode, driven by the one signal that observably announces it, is the transition the policy does not perform. The rollout says the policy is bad; it does not say *why*. The policy might ignore force entirely; it might respond to force but too weakly; it might respond in a form that collapses precisely when force exceeds the demonstrated range. These hypotheses have opposite prescriptions (more data? different sensors? different architecture?), and none of them is distinguishable from training curves or rollouts. Distinguishing them requires intervening on the signal itself—the instrument we build next.

IV. AN INSTRUMENT FOR CAUSAL AUTHORITY

The architectural half of our recipe doubles as an instrument. It has two parts: a conditioning pathway constructed so that interventions on it are meaningful (IV-A–IV-B), and a battery of counterfactual probes that perform the interventions (IV-C). Throughout, the design goal is *measurement validity*, not architectural novelty: every component exists to make one number well-defined—how much does the condition causally move the action?

A. Computed condition vector $\hat{c}$

From the wrench F and seal bit already present in the naive policy’s state we compute a small set of task-critical conditions—the physical cues a human operator references (“if you feel contact, stop; if the seal is on, lift”):

$$\begin{aligned}\hat{c}_{\text{contact}} &= \text{clip}((\|F\| - F_0) / \tau, 0, 1), \\ \hat{c}_{\text{fmag}} &= (\|F\| - F_0) / \tau_m \quad (\text{unsaturated}), \\ \hat{c}_{f_z} &= (f_z - o_z) / \tau_z, \quad \hat{c}_{\text{seal}} = \text{seal},\end{aligned}\tag{1}$$

with offsets and scales calibrated to the measured force statistics of the demonstrations (e.g., F_0 at the contact-onset magni-

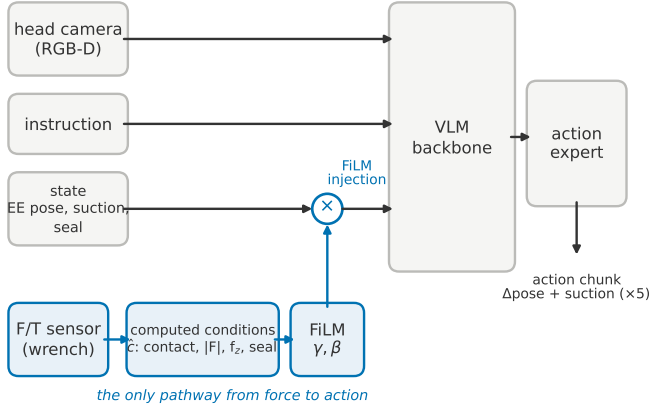


Fig. 1. The conditioned policy. The raw wrench does not enter the policy state; force reaches the action only as the computed condition vector $\hat{c}$, injected through FiLM modulation of the state token entering the VLM backbone. Because this pathway is explicit and unique, forcing $\hat{c}$ at inference measures the causal authority of the condition.

tude).¹ Two properties are essential. *Computed, not learned*: $\hat{c}$ is a fixed function of observed signals, so its physical meaning cannot drift during training—which is what makes forcing $\hat{c}$ at inference a well-defined intervention rather than a perturbation of an arbitrary latent. *Zero new information*: every channel is a deterministic function of state dimensions the naive policy also receives, so conditioning changes *routing*, never *content*—the “it works because it got extra sensing” objection is excluded by construction. The channels are deliberately plural: keeping contact, force magnitude, and seal separate is what later lets the probes attribute authority to specific signals (Sec. V-B).

B. FiLM injection and the force-mask bottleneck

$\hat{c}$ enters the policy through FiLM modulation [7], with the modulation initialized to identity so that training starts from the unconditioned policy; the injection point is the state token entering the VLM backbone.² With $\hat{c}$ injected but the raw wrench still present in the state, the policy has two routes to force. Masking the wrench dimensions from the state closes the raw route, making $\hat{c}$ the *only* pathway from force to action. The mask is the instrument’s validity device: under it, forcing $\hat{c}$ intervenes on the totality of the policy’s force access, so measured authority is a property of the policy’s force use, not of one redundant pathway among several. It also strictly *reduces* the conditioned policy’s information relative to the naive policy. The unmasked configuration is retained as a control—and turns out to demonstrate bypass (Sec. V-A).

¹Calibration quality matters and is itself a finding. The conditioned policies in this paper are separately trained instances of the same design: the bypass matrix (Sec. V-A) and the robot picks (Sec. V-E) use an earlier calibration; the recipe ablation (Sec. V-C) and the live counterfactuals (Sec. V-F) use offsets re-estimated from the measured contact distribution plus the unsaturated $\hat{c}_{\text{fmag}}$ channel. An uncalibrated transplant onto a different architecture fails outright (Sec. VI).

²Injecting at the action-expert input instead acquired consistently less authority in development; we do not pursue it here.

C. Counterfactual probe battery

All probes intervene on a trained, frozen policy; none involves retraining. We define **authority** as the intervention-induced change in the commanded vertical action on committed-descent frames, expressed as a percentage of the mean commanded descent: 0% means the intervention is ignored, 100% means it fully cancels descent, >100% means reversal (retreat).

(P1) Condition forcing (open loop). With images and state frozen, $\hat{c}$ is forced from its observed value to a target pattern—either all channels saturated (*full forcing*) or a calibrated contact-onset pattern matching measured first-contact statistics (*contact-moment forcing*). Applicable to conditioned models only.

(P2) State transplant (open loop). The force-bearing dimensions of a *measured* first-contact state are transplanted into pre-contact descent frames of held-out episodes. This probe applies to any policy, including the naive policy, and supports a pathway decomposition: apply the transplant to the raw state only (d_{raw}), to the $\hat{c}$ computation only (d_{FiLM}), or to both (total), attributing the response to the raw route versus the conditioning route. **[TODO: verify exact transplanted dims (wrench+seal vs full state) in probe_state_authority.py before submission]**

(P3) Dose-response sweep. The transplanted wrench is rescaled to target magnitudes (8, 10, 12 N), tracing the policy’s response as a function of force *dose*. The demonstrations peak near the low end of this range, so the sweep measures extrapolation: a policy that has bound braking to force should brake *more* at higher dose; one that has memorized a contact template should respond most near the template and fade beyond it. The sweep’s *slope sign* is the paper’s central discriminator.

(P4) Closed-loop press simulation. Open-loop probes cannot show consequences. We place the frozen policy in a closed loop on held-out episodes: commanded descent advances a virtual penetration depth, a linear-stiffness force model raises the simulated wrench accordingly, and the seal event is withheld—simulating the operationally critical misalignment scenario in which no press depth ever produces a seal. Outcomes: whether the policy stops before the force limit, and maximum penetration. A control variant grants the seal at a shallow depth, separating force-driven stopping from seal-event-driven stopping.

(P5) Live counterfactual probe. The full loop contains physics no offline probe reproduces. On the robot, we freeze observations at staged poses along the approach and apply (P1)–(P3) interventions to *two* policies—the conditioned policy and the naive policy—on identical frozen observations, so both receive the same physical counterfactual. Two operational details matter for validity: force/torque baseline drift is re-anchored at a pre-descent hover pose before probing (drift of up to ~ 1 N within a day was measured, pose-sensitive), and absolute transplant targets are shifted by the measured drift so that both models receive the same effective dose.

TABLE I

BYPASS MATRIX: VALIDATION LOSS VS. CAUSAL AUTHORITY. FULL FORCING = P1 ALL-CHANNEL; CONTACT-MOMENT = P1 CALIBRATED ONSET PATTERN.

Policy	Val loss	Authority	
		full forcing	contact-moment
naive VLA (full wrench)	0.15087	—	—
FiLM, wrench kept	0.15075	7%	3%
FiLM, wrench masked	0.14624	76%	7%

Scope. P1–P3 are single-frame and open-loop; P4 closes the loop under a force model; P5 closes the gap to the real system. No single probe is decisive—the findings below rest on their agreement.

V. EXPERIMENTS

We present the offline probe battery first (V-A–V-D), then the robot validation (Sec. V-E–V-F). All offline probes run on the six held-out validation episodes, and every reported model is its validation-loss-selected (*best*) checkpoint—one uniform selection rule for all policies; the robot deployments below use the same checkpoints.

A. Bypass: standard validation cannot see causal use

Table I varies wrench masking for FiLM-conditioned policies trained identically on the same data.

Two observations. **(i) Bypass.** With the wrench left in the state, the FiLM-conditioned policy trains to a validation loss equal to the naive policy’s in the *fourth decimal place* (0.15075 vs. 0.15087) while its condition channels carry 7% authority: gradient descent satisfied the imitation objective through the raw route and left the dedicated conditioning pathway causally dead. Nothing in the training signal distinguishes this policy from one that uses its conditions. **(ii) The mask reroutes, at no cost.** Masking the wrench makes the same conditioning pathway carry 76% authority—and yields the *lowest* validation loss of the three policies: closing the raw route costs nothing on the manifold. The bypass result replicates in an independent pair below (Sec. V-C): variants with and without the mask again differ by 10^{-4} in loss and categorically in authority.

B. Where authority binds: stable states, not the transient

Authority, where it exists, is not uniformly distributed over the condition channels. Contact-moment forcing (P1, calibrated onset pattern) yields only 7% even for the best-conditioned model (Table I, last column): the 1–2-frame contact *transient* is weakly adopted across the board. The state-transplant probe (P2) shows what is adopted instead. Transplanting a measured first-contact state into pre-contact descent frames moves the *naive policy* by 105% of its descent command—the naive policy is not force-blind; its raw route responds strongly to force-bearing state. But transplanting a *sealed* state moves both models far more than the contact state does (on-robot: 147% vs. 67% for the naive policy; Sec. V-F), and the strongest single responses everywhere are to the stable,

[TODO: dose-response figure: cancellation vs. forced dose (8/10/12 N) for the four ablation policies — assets: vla/probes/0729_state_*_ramp*.txt]

Fig. 2. Dose-response under forced wrench magnitude (P3). The naive policy’s response fades past the demonstrated range; the unmasked variant lies on top of it (all response through the raw route); the ungrounded bottleneck is flat at zero; only the grounded bottleneck grows with dose, crossing full cancellation into commanded retreat at 12 N.

sustained signals—seal and sustained press—rather than to contact onset.

The picture: what a policy adopts is decided by the *statistical character* of the signal. Stable, persistent conditions (seal state) are adopted essentially for free; a transient that is redundant with depth-like cues in ordinary demonstrations is not. This has an operational corollary developed in Sec. VI: on stiff contact, even a policy with substantial condition authority cannot succeed by *reactive* gating at the contact moment—the successful mechanism is anticipatory behavior plus post-hoc state gating, which is exactly what the probes measure as adopted.

C. Ablating the recipe: each ingredient is necessary

The section’s central experiment removes the recipe’s ingredients one at a time while holding everything else fixed. Starting from the *same* trained naive checkpoint, on the *same* demonstrations, we continue training four policies: the naive policy itself; FiLM *without* the mask (removes the bottleneck’s exclusivity); FiLM with mask but *shuffled* $\hat{c}$ during training (removes the grounding: the bottleneck exists, its content is noise); and the full recipe. Imitation error on held-out episodes is indistinguishable across all four (0.84–0.96 mm per step): on the data manifold, these are the same policy. Under intervention they are three different mechanisms (Table II, Fig. 2).

Grounding, not capacity, is the active ingredient. The ungrounded control kills every alternative explanation at once: it has the same added parameters, the same continued training, the same bottleneck—and zero response at every dose, with the worst closed-loop outcome in the study (431 mm penetration). An ungrounded bottleneck is *worse* than raw access: the mask removed the raw route and the noise-trained $\hat{c}$ replaced it with nothing. Conversely, the unmasked control shows that adding the FiLM module without the mask changes nothing: its dose-response curve lies on the naive policy’s (+1.57 → +1.12 vs. +1.41 → +1.12; identical at 12 N), its pathway decomposition puts the response entirely on the raw route, and its loss matches the grounded variant’s to the fourth decimal (0.14750 vs. 0.14762)—the bypass finding, replicated with authority now at 94% vs. 0.

The form dissociation. The naive policy responds to force—at the demonstrated magnitude. As the dose rises past the demonstrations, its response *fades* (36% → 28%): a template, matched ever more poorly as reality departs from it. The grounded bottleneck’s response *grows monotonically*, crossing full cancellation into retreat at 12 N—where the unsaturated

TABLE II

RECIPE ABLATION: SAME DATA, SAME INITIALIZATION, INDISTINGUISHABLE IMITATION—THREE MECHANISMS. DOSE-RESPONSE = P3 CANCELLATION AT 8 N→12 N (% OF OWN DESCENT). PRESS SIM = P4 SEAL-NEVER: EPISODES STOPPED BEFORE FORCE LIMIT / MAX PENETRATION.

Policy	Force access	Dose-response 8→12 N	Transplant (P2)	Press sim (P4)	Val err
naive VLA	raw wrench	+1.41 → +1.12 mm (36→28%) ↓	+1.63 (105%)	4–5/6, max 14.9 mm	0.84 mm
FiLM, wrench kept	raw + dead $\hat{c}$	+1.57 → +1.12 mm (d_{raw} carries all)	+1.68 ($d_{\text{FiLM}}=0.04$)	4–5/6, max 11.9 mm	0.85 mm
FiLM, mask, shuffled $\hat{c}$	ungrounded bottleneck	−0.06 → −0.11 mm (≈ 0)	−0.06 (0%)	0/6, max 431 mm	0.96 mm
FiLM, mask, grounded $\hat{c}$	grounded bottleneck	+1.20 → +4.40 mm (31→ 115%) ↑	+1.57 (94%, all d_{FiLM})	6/6, max 4.5 mm	0.87 mm

channel reads five times its training range. The monotone extrapolation is not learned from data (no demonstration presses to these forces); it is inherited from the channel’s design as a calibrated monotone scalar. Same data, same imitation: raw access binds a template, the grounded bottleneck binds a brake.

Checkpoint robustness, and what the naive policy’s stopping is bound to. The forms are not artifacts of checkpoint selection: no naive checkpoint we examined exceeds 65% cancellation at any dose—later checkpoints respond more in open loop yet fare *worse* in the closed-loop press simulation, because a response that saturates below full cancellation cannot stop a stiff press regardless of its size—while the grounded brake remains monotone to the end of training. Finally, the seal-granted control variant of P4 localizes the naive policy’s failure: given a seal event at shallow depth it stops normally (0.5 mm over-travel); withhold the seal and it presses through. Its stopping is real but bound to the *seal event*; the grounded policy’s is bound to *force*.

D. Scale does not change the class

[TODO: HOLD (08-11): pi0 offline re-run in progress — replace the numbers in this subsection when it lands (RESULTS_LINEUP A3)] Fine-tuning $\pi_{0.5}$ (3.6B parameters, $8 \times \text{SmolVLA}$) [24] on the same demonstrations reproduces the naive-policy signature: strong on-manifold transplant response (65%), a dose-response that saturates shallowly (never exceeding 37% cancellation, approaching no full stop at any dose), and closed-loop collapse in the seal-never press simulation (0/6, max penetration 86 mm—worse than the smaller model), at equal-or-better imitation error. Neither parameter count nor a different VLA architecture converts template responding into force-bound braking; the failure class is a property of the objective-plus-data, not of model scale.

E. Robot validation: closed-loop picks

The offline battery makes three predictions about the real closed loop. A template response that fades beyond the demonstrations should let contact force run away until something outside the policy intervenes; a monotone brake should stop on its own, at forces near the demonstrated range; and the two responses should cross between the demonstrated magnitudes and the safety limit. The robot experiments test these predictions, and the outcome variable is the *force behavior* of each attempt—task success follows from it by definition.

The predictions hold (Fig. 3). Under the naive policy every attempt ends the same way: contact, continued descent, and

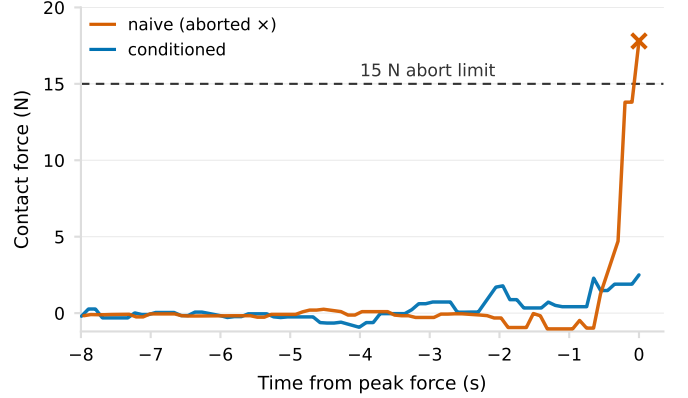


Fig. 3. Contact force on the robot: one representative trial per policy, aligned at peak force ($t=0$), over the interaction window (before lift, where carrying the case would re-enter payload weight into the measured force). The naive attempt rises through the demonstrated range and is terminated by the external 15 N abort ($\times$; all three naive attempts end this way, at 15.4–18.7 N), while the conditioned pick presses to 2.5 N—the median of its nine picks across three stack heights, one of them absent from the demonstrations—and stops on its own.

an external force-limit abort at 15.4–18.7 N—the closed-loop realization of a response that saturates below full cancellation. The conditioned policy stops itself in *every* pick, across three stack heights: 3/3 at the naive policy’s own height, and 3/3 at an intermediate height absent from the demonstrations, which contain only the other two. Peak force, measured over the *interaction window* (descent through retreat, before lifting re-enters payload weight into the wrench; the abort monitors the same quantity), stays at or below 3.1 N in seven of the nine picks (median 2.5 N; two picks sealed with no measurable compression at all). The two force distributions do not overlap, and the task outcomes are their threshold readout: 0/3 versus 9/9. The two harder presses (10.2 and 14.5 N) are the brake visible near its limit—both stopped short of the abort threshold and completed the pick where the naive policy, at the same forces, was still commanding descent.

The stopping runs through the bottleneck. The deployed policy’s own dose-response is monotone: transplanted wrenches at 8/10/12 N cancel 3% → 7% → 13% of committed descent—rising with dose, and carried entirely by the FiLM path (the masked raw route contributes +0.00 mm, the mask working as designed). During the picks themselves the modulation is measurably active at the transitions ($|\gamma|$ +48%, $|\beta|$ +79% at contact/seal relative to descent), and forcing $\hat{c}$ at ten frozen

[TODO: live counterfactual figure: fair-dose bars + high-dose crossover curves (assets: vla/live_film_probes 08-06 runs, _vs_naive.png)]

Fig. 4. On-robot counterfactuals (P5), naive vs. recalibrated conditioned policy on identical frozen observations. Left: at demonstration-level doses the naive response is equal or larger. Right: raising the dose to +6/+9/+12 N reverses the ordering between 6 and 9 N—the offline crossover, reproduced on the physical system.

poses on the robot moves the commanded action in the correct direction at every staged condition (contact +0.73, sealed +2.09, +6 N +2.18 mm/frame).

F. Robot validation: on-robot counterfactuals

The live probe (P5) freezes observations at staged poses and doses the naive policy and the conditioned policy with identical counterfactual states. Two sessions matter (Fig. 4).

Fair dose. At transplant doses matching the demonstrations ($\leq$ contact magnitude), the *naive* policy responds as strongly or stronger than the conditioned policy (first-contact: 67% vs. 32% of own descent; sealed state: 147% vs. 58%)—in quantitative agreement with the offline dose-response curves at the same doses, where the template peaks. The conditioned policy’s response is dose-monotone across the staged conditions (hover +0.29 $\rightarrow$ pre-seal +0.87 $\rightarrow$ sealed +1.41 $\rightarrow$ +6 N +1.70 mm/frame).

High dose. Raising the forced normal force to +6/+9/+12 N reproduces the offline crossover on the robot: the conditioned policy accelerates its braking, +1.19 $\rightarrow$ +4.59 $\rightarrow$ +8.56 mm/frame—at 12 N, twice its own descent command, i.e., commanded retreat—while the naive policy’s response collapses at the margin (+1.34 $\rightarrow$ +3.18 $\rightarrow$ +4.18; increments +1.84 $\rightarrow$ +1.00). The reversal sits between 6 and 9 N. At near-surface poses the separation is starkest: naive +1.0–1.8 vs. conditioned +7.5 mm/frame at 12 N.

Isn’t the naive policy better, then? At mid dose, yes—and it does not matter. Over-press is a dynamical process in which force rises *past* the demonstrated magnitude; what determines the outcome is the response exactly where the template fades. The naive policy’s brake weakens as it becomes more necessary (closed loop: force runaway in every attempt; press-simulation penetration to the force limit), while the conditioned brake strengthens (self-stop in every pick; 6/6 simulated presses converging). The naive policy’s strongest counterfactual response is to the *sealed* state—but over-press is precisely the scenario in which the seal never arrives: its most powerful brake is bound to an absent signal. Similar in loss, similar at the demonstrated doses, categorically different where the task is decided.

Three validity notes. Force-torque baselines were re-anchored at a pre-descent hover before each probe session, and absolute transplant targets were drift-shifted so both policies received the same physical dose; live conditioned-policy magnitudes are conservative lower bounds, as some frozen states are slightly off-manifold for its input distribution. The naive policy’s closed-loop attempts were run at a single

stack height—its failure mode is uniform and force-driven, but the height-generalization evidence applies to the conditioned policy only. High-dose probes cover three poses; we report them as targeted supplementary measurements, not a survey.

VI. DISCUSSION AND LIMITATIONS

Why the prescription must be anticipatory. A first-principles calculation explains both the failure and the fix. The case stack is stiff: ~ 5 N/mm of penetration. Stopping below the 15 N limit therefore requires a full stop within ~ 2 mm of contact—under 330 ms action chunks and the demonstrated descent rates, this is arithmetically impossible for *reactive* gating that begins braking at the contact transient, no matter how much authority the transient carries. The viable mechanisms are anticipatory: approach slowly, press deliberately, retreat. The press-retreat demonstrations teach exactly this behavior prior, and the probes confirm the resulting division of labor: the successful policy’s contact-moment authority is small (7%) while its full-condition authority is large (76%)—success comes from an anticipatory behavior prior plus post-hoc state gating, not from a reflex. The instrument’s value here is not that it found a large number, but that it correctly *predicted which mechanism could not work*.

Grounding is not free. Transplanting the instrument naively fails, and the failure is diagnostic. Attaching the same conditioning to $\pi_{0.5}$ via action-expert injection with uncalibrated offsets produces the ungrounded-control signature at scale: zero authority at every dose—including with the contact channel forced to saturation—and closed-loop penetration *worse* than the raw $\pi_{0.5}$ baseline (0/6, max 94–102 mm vs. 86 mm), at unchanged imitation error. The mask removes the raw route; if $\hat{c}$ fails to ground, what remains is a policy with genuinely no force pathway. A bottleneck plus counterfactual data is necessary but not sufficient: injection site and calibration must be re-engineered per architecture (consistent with the calibration sensitivity observed within SmolVLA). [DECISION: keep this paragraph’s numeric detail vs. compress to two sentences — page budget]

Operational asymmetry. The computed condition creates knobs the baseline lacks. Force-torque baselines drift (~ 1 N within a day, pose-sensitive); the conditioned policy re-anchors its offsets at a hover pose each run—a deployment-time correction with no retraining—while the baseline’s force statistics are frozen in its normalization. The same interface is what makes the policy auditable (P1/P5 probes) and intervenable at deployment.

Limitations. Single task, single platform, small robot- n : the robot results are a case study, and the statistical claims live in the offline battery ($n=6$ held-out episodes $\times$ dense frames, five matched policies, two architectures). The press simulation depends on a linear force model; we mitigate with the seal-granted control and the live probes, which bracket the simulated mechanism from both sides. Contact-moment (reactive) authority plateaus low in every configuration we trained—in development phases, oversampling transition frames also failed to move it; distilling an explicit descend-until-contact

oracle (e.g., via DAgger [20], [25]) is the natural next step, as is a negative-control task whose critical condition is *not* observable in the state, and replication of the bypass audit on force-fusion architectures whose designs assume use [1], [3].

VII. CONCLUSION

On a task whose critical transition is observable only through force, we showed that a behavior-cloned VLA policy given the full signal does not bind its behavior to it—and that no number in the standard validation stack could have revealed this. A computed condition vector behind a zero-new-information FiLM bottleneck turns the question into a measurement: counterfactual authority probes expose bypass at identical loss, localize authority to stable post-hoc states rather than the contact transient, and—in a matched quintuple with indistinguishable imitation error—show that the data’s correlation structure decides *whether* force is used while the conditioning design decides the *form* its use takes: a template that fades beyond the demonstrations, or a calibrated brake that strengthens monotonically, stops, and retreats. The distinction is invisible on the manifold and decisive off it, on the robot as offline. Imitation pins down the trajectory. If we want the mechanism pinned too, we must give the policy demonstrations that decorrelate the condition from its correlates—and an interface through which the condition’s use can be shaped, measured, and corrected.

REFERENCES

- [1] TODO-verify authors, “ForceVLA: Force-aware mixture-of-experts for contact-rich manipulation,” *arXiv preprint arXiv:2505.22159*, 2025.
- [2] Wang et al. (TODO-verify), “PhaForce: Phase-scheduled visual-force policy learning with slow planning and fast correction,” *arXiv preprint arXiv:2603.08342*, 2026.
- [3] TODO-verify authors, “ForceFlow (todo-verify full title),” *arXiv preprint arXiv:2605.11048*, 2026.
- [4] —, “FILIC: Dual-loop force-guided imitation learning with impedance control (todo-verify),” *arXiv preprint arXiv:2509.17053*, 2025.
- [5] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” in *NeurIPS*, 2019, arXiv:1905.11979.
- [6] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [7] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *AAAI*, 2018.
- [8] Z. Zhang, H. Xu, Z. Yang, C. Yue, Z. Lin, H. ang Gao, Z. Wang, and H. Zhao, “TA-VLA: Elucidating the design space of torque-aware vision-language-action models,” in *CoRL*, 2025, arXiv:2509.07962.
- [9] TODO-verify authors, “FARM: Tactile-conditioned diffusion policies (todo-verify full title),” *arXiv preprint arXiv:2510.13324*, 2025.
- [10] —, “TacForeSight: Force-conditioned tactile world models (todo-verify full title),” *arXiv preprint arXiv:2606.11184*, 2026.
- [11] J. Huang, S. Wang, F. Lin, Y. Hu, C. Wen, and Y. Gao, “Tactile-VLA: Unlocking vision-language-action model’s physical knowledge for tactile generalization,” *arXiv preprint arXiv:2507.09160*, 2025.
- [12] S. Nasiriany et al., “RT-Affordance: Affordances are versatile intermediate representations for robot manipulation,” *arXiv preprint arXiv:2411.02704*, 2024.
- [13] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, “Fighting copycat agents in behavioral cloning from observation histories,” in *NeurIPS*, 2020, arXiv:2010.14876.
- [14] K. Hermann, H. Mobahi, T. Fel, and M. C. Mozer, “On the foundations of shortcut learning,” *arXiv preprint arXiv:2310.16228*, 2023.
- [15] TODO-verify authors, “Shortcut learning in generalist robot policies (todo-verify full title),” *arXiv preprint arXiv:2508.06426*, 2025.
- [16] —, “Intervention-based evaluation of imitation policies (todo-verify full title),” *arXiv preprint arXiv:2507.22380*, 2025.
- [17] —, “Interventional imitation learning (todo-verify full title),” *arXiv preprint arXiv:2307.15980*, 2023.
- [18] J. Jones, O. Mees, C. Sferrazza, K. Stachowicz, P. Abbeel, and S. Levine, “Beyond sight: Finetuning generalist robot policies with heterogeneous sensors via language grounding,” in *ICRA*, 2025, arXiv:2501.04693.
- [19] R. Li, Q. Li, R. Ma, Y. Deng, L. Luo, Z. Du, J. Xiang, H. Liang, R. Wang, J. Yang, and B. Guo, “FM-VLA: Force-based memory for vision-language-action models in contact-rich manipulation,” *arXiv preprint arXiv:2607.18231*, 2026.
- [20] D. Chen, B. Zhou, V. Koltun, and P. Krähenbühl, “Learning by cheating,” in *CoRL*, 2019.
- [21] TODO-verify authors, “Learning long-horizon robot manipulation skills via privileged action,” *arXiv preprint arXiv:2502.15442*, 2025.
- [22] J. A. Collins, C. Houff, Y. L. Tan, and C. C. Kemp, “ForceSight: Text-guided mobile manipulation with visual-force goals,” in *ICRA*, 2024, arXiv:2309.12312.
- [23] M. Shukor et al., “SmolVLA: A vision-language-action model for affordable and efficient robotics,” *arXiv preprint arXiv:2506.01844*, 2025, tODO-verify arXiv ID.
- [24] K. Black et al., “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” *arXiv preprint arXiv:2504.16054*, 2025, tODO-verify arXiv ID and author list.
- [25] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *AISTATS*, 2011.